



**BAULKHAM HILLS HIGH SCHOOL**

**2015**

**YEAR 12 HALF-YEARLY**

# Mathematics Extension 1

## General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

**Total marks – 70**

**Exam consists of 8 pages.**

This paper consists of TWO sections.

**Section 1 – Pages 2-3**

**Multiple Choice**

Question 1-10 (10 marks)

**Section 2 – Pages 4-7**

**Extended Response**

Question 11- 14 (60 marks)

**Standard integrals provided on page 8**

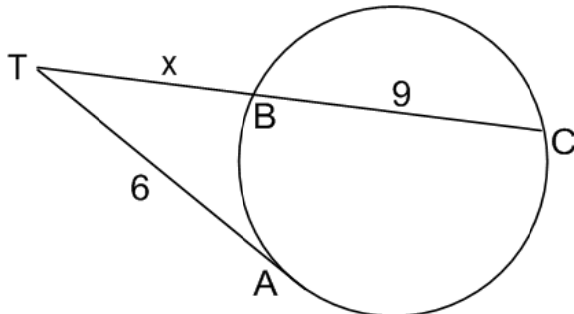
**Section I – 10 marks**

**Attempt Questions 1 to 10**

**Allow about 15 minutes for this section.**

**Use the multiple choice answer sheet provided in the answer booklet.**

- 1 TA is a tangent to the circle at A and TC is a secant, meeting the circle at B.



Given that  $TA=6$ ,  $CB=9$  and  $TB=x$ , what is the value of  $x$ ?

- (A) -12 (B) 2 (C) 3 (D) 4

1

- 2 Let  $\alpha, \beta, \gamma$  be the roots of  $x^3 - 4x + 1 = 0$ . What is the value of  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ ?

- (A) -4 (B) -1 (C) 1 (D) 4

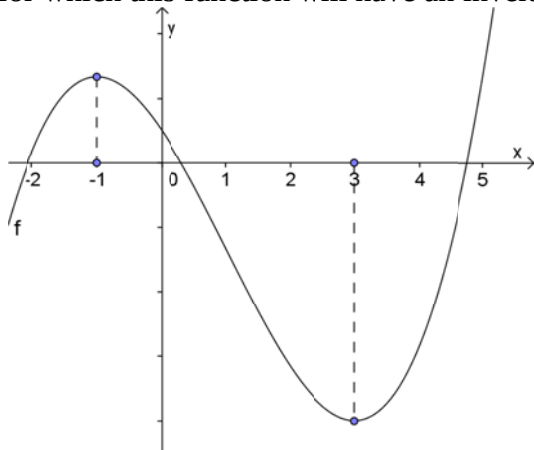
1

- 3 Two of the roots of the equation  $x^3 - 2x^2 + kx + 18 = 0$  are equal in magnitude but opposite in sign. What is the value of  $k$ ?

- (A) -9 (B) -6 (C) 6 (D) 9

1

- 4 The diagram shows a polynomial function  $y = f(x)$ . What is the largest possible domain containing  $x = 0$  for which this function will have an inverse function?



- (A)  $x \leq 1$  (B)  $-1 \leq x \leq 3$  (C)  $0 \leq x \leq 3$  (D)  $x \leq 0$

1

- 5 Given the coordinates of A and B are  $(-1, 1)$  and  $(3, -1)$  respectively, find the coordinates of the point P which divides AB externally in the ratio 1:3.

- (A)  $(2, -\frac{1}{2})$  (B)  $(-\frac{3}{2}, \frac{1}{2})$  (C)  $(-3, 2)$  (D)  $(4, -2)$

1

- 6 The number of different arrangements of the word SIMILES which begin and end with letter S is:

(A)  $\frac{7!}{2!}$  (B)  $\frac{7!}{2!2!}$  (C)  $\frac{5!}{2!}$  (D)  $\frac{5!}{2!2!}$

1

7  $\int 6 \cos^2 3x \, dx =$

(A)  $2 \sin^3 3x + c$  (B)  $2 \cos^3 6x + c$   
(C)  $3x + \frac{1}{2} \sin 6x + c$  (D)  $3x - \frac{3}{2} \sin 6x + c$

1

- 8 A curve is defined by the parametric equations  $x = \sin 2t$  and  $y = \cos 2t$ . Which of the following is equal to  $\frac{dy}{dx}$ ?

(A)  $\cos 4t$  (B)  $2 \tan 2t$  (C)  $2 \sin 4t$  (D)  $-\tan 2t$

1

- 9 If  $y = [\ln(4x - 1)]^2$  then  $\frac{dy}{dx}$  is equal to:

(A)  $\frac{8 \ln(4x-1)}{4x-1}$

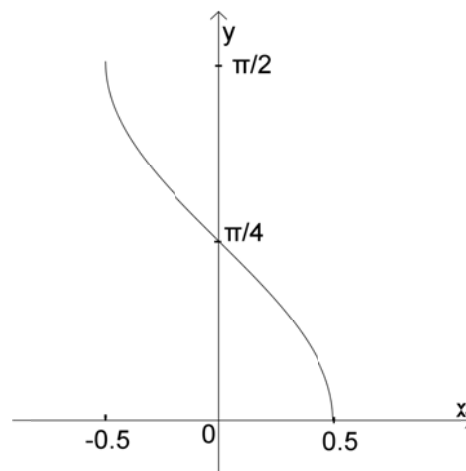
(B)  $\frac{2 \ln(4x-1)}{4x-1}$

(C)  $\frac{8}{4x-1}$

(D)  $\frac{2}{4x-1}$

1

10



The graph above shows:

(A)  $y = 2 \cos^{-1} 2x$  (B)  $y = \frac{1}{2} \cos^{-1}(\frac{1}{2}x)$   
(C)  $y = 2 \cos^{-1}(\frac{1}{2}x)$  (D)  $y = \frac{1}{2} \cos^{-1} 2x$

## Section II – Extended responses

Attempt all questions. Show all necessary working.

Answer each question in the appropriate pages of your answer booklet.

### Question 11 (15 marks)

- a) (i) Express  $\sqrt{3}\cos x - \sin x$  in the form  $A\cos(x + \alpha)$  where  $A > 0$  and  $0 < \alpha < \frac{\pi}{2}$ . 2
- (ii) Without using calculus, find the maximum value of  $3 - 3\cos x + \sqrt{3}\sin x$ . 2
- b) Given  $P(x) = -x^3 + 3x + 2$ :
- (i) Show that  $x + 1$  is a factor of  $P(x)$  1
- (ii) Fully factorise  $P(x)$  2
- (iii) Sketch the graph of  $y = P(x)$  2
- c) 5 men and 3 women are candidates for election to a committee of 4 people. How many committees are possible if there must be at least one man and at least one woman selected? 2
- d) Given  $f(x) = \tan^{-1} \frac{4}{x} + \tan^{-1} \frac{x}{4}$ :
- (i) find  $f'(x)$  2
- (ii) sketch  $y = f(x)$  2

### Question 12 (15 marks)

- a) Without the use of a calculator, find the exact value of:  
 $\sin(2 \tan^{-1} \sqrt{2})$  2
- b) For the lines  $2x - y + 1 = 0$  and  $x - 4y + 4 = 0$
- (i) Find the acute angle between the lines, correct to the nearest degree. 2
- (ii) Find the area of the triangle bounded by these two lines and the  $x$ -axis. 2
- c) Solve the equation:  $\sin 2x = \cos x$  for  $0 \leq x \leq 2\pi$  2

*Question 12 continues on the next page*

**Question 12 (cont)**

- d) Solve:  $\ln(2x^2 + 3x - 9) = \ln(2x^2 - 7x + 6) + 2$  3  
expressing your answer in terms of e.
- e) Given  $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ ,  
find  $\int 2\sin^3\theta \, d\theta$  2
- f) Simplify :  $\sin(a+b)\cos\left(\frac{\pi}{2} - a - b\right) - \cos(a+b)\cos(\pi - a - b)$  2

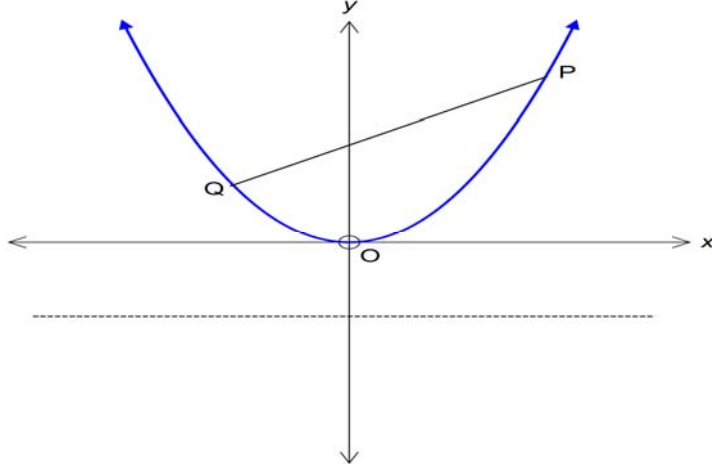
**Question 13 (15 marks)**

- a) (i) Prove by Mathematical Induction that  $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$   
for all positive integers n 3
- (ii) Hence evaluate  $11^3 + 12^3 + 13^3 + \dots + 50^3$  1

**Question 13 continues on the next page**

### Question 13 (cont)

- b) The points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$ .



The equation of the chord  $PQ$  is  $y = \left(\frac{p+q}{2}\right)x - apq$  (Do NOT prove this)

- (i) Show that if  $PQ$  passes through the focus  $S(0, a)$  then  $pq = -1$  1
  - (ii)  $X$  is the midpoint of the focal chord  $PQ$ .  $T$  lies on the directrix such that  $XT$  is perpendicular to the directrix and  $W$  is the midpoint of  $XT$ . Find the equation of the locus of  $W$ . 3
- c) Newton's Law of Cooling states that the rate of change of the temperature ( $T$ ) of a body is proportional to the difference between the temperature of the body and the temperature of the surrounding medium ( $P$ )  
 ie  $\frac{dT}{dt} = k(T - P)$
- (i) If  $A$  and  $k$  are constants, show that  $T = P + Ae^{kt}$  satisfies Newton's Law of Cooling. 1
  - (ii) A cup of coffee is made with temperature  $100^\circ\text{C}$ . Two minutes later, its temperature is  $93^\circ\text{C}$  but it is still too hot to drink. If the temperature in the room is  $23^\circ\text{C}$ , evaluate  $A$  and  $k$  (correct to 3 significant figures) 2
  - (iii) The coffee is drinkable when it reaches  $80^\circ\text{C}$ . How many minutes after the coffee is made will this be? 1

- d) Solve

$$\frac{x^2+x-6}{x} \geq 0$$

3

**Question 14 (15 marks)**

a) Find  $\int \tan^2 2x \, dx$  2

b) Simplify  $\frac{2^{4n} \times 3^{2n}}{8^n \times 6^n} + 3^n$  2

c) A function is defined by  $f(x) = x - \frac{1}{x}, x > 0$

(i) Show that the inverse function is given by  $f^{-1}(x) = \frac{x}{2} + \sqrt{1 + \frac{x^2}{4}}$  2

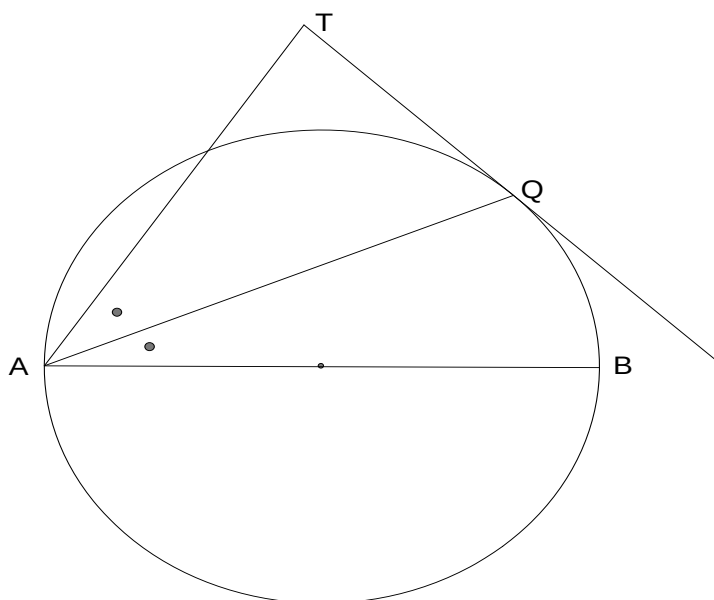
(ii) Sketch the curves  $y = f(x)$  and  $y = f^{-1}(x)$  on the same set of axes 2

(iii) State the domain and range of  $f^{-1}(x)$  2

d) Prove that  $\frac{\sin 8x}{1 + \cos 8x} = \tan 4x$  2

- e) In the diagram, AB is a diameter of the circle.  
TQ is a tangent and AT is a secant which meets the circle at P.  
AQ bisects  $\angle BAT$ .

Copy the diagram into your answer booklet.



3

Prove that  $\angle ATQ = 90^\circ$ .

**End of Examination**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

## Section I.

$$\begin{aligned}
 1. \quad x(x+9) &= 36 \\
 x^2 + 9x - 36 &= 0 \\
 (x-3)(x+12) &= 0 \\
 x &= 3, -12 \quad (C)
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} & \quad [x^3 + 10x^2 - 4x + 1] \\
 &= \frac{-4}{-1} = +4 \quad (D)
 \end{aligned}$$

$$\begin{aligned}
 3. \quad \text{Let roots} &= \alpha, -\alpha, \beta. \\
 \alpha - \alpha + \beta &= 2 \quad \therefore \beta = 2 \\
 \alpha\beta - \alpha\beta - \alpha^2 &= k \quad \therefore k = -\alpha^2 \\
 \alpha \cdot -\alpha \cdot \beta &= -18 \\
 -\alpha^2 \cdot 2 &= -18 \\
 \therefore k = -\alpha^2 &= -9 \quad (A)
 \end{aligned}$$

$$4. \quad -1 \leq x \leq 3 \quad (B)$$

$$\begin{aligned}
 5. \quad &\left( \frac{1(3) - 3(-1)}{1-3}, \frac{1(-1) - 3(1)}{1-3} \right) \\
 &= (-3, 2) \quad (C)
 \end{aligned}$$

$$\begin{aligned}
 6. \quad &S \text{ --- } S \\
 &5 \text{ letters, 2 same.} \\
 &\frac{5!}{2!} \quad (C)
 \end{aligned}$$

$$\begin{aligned}
 7. \quad &6 \int \cos^2 3x \cdot dx \\
 &= 6 \int \frac{1}{2} (1 + \cos 6x) \cdot dx \\
 &= 3 \int 1 + \cos 6x \cdot dx \\
 &= 3 \left( x + \frac{1}{6} \sin 6x \right) + c \\
 &= 3x + \frac{1}{2} \sin 6x + c \quad (C)
 \end{aligned}$$

$$\begin{aligned}
 8. \quad \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} \\
 &= \frac{-2 \sin 2t}{2 \cos 2t} \\
 &= -\tan 2t \quad (D)
 \end{aligned}$$

$$\begin{aligned}
 9. \quad \frac{dy}{dx} &= 2 \ln(4x-1) \cdot \frac{1}{4x-1} \cdot 4 \\
 &= \frac{8 \ln(4x-1)}{4x-1} \quad (A)
 \end{aligned}$$

$$10. \quad \text{Ampl. halved} \quad (D)$$

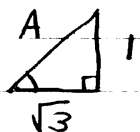
# Question 11

$$a) (i) \sqrt{3} \cos x - \sin x = A \cos(x + \alpha)$$

$$= A \cos x \cos \alpha - A \sin x \sin \alpha$$

$$A \cos \alpha = \sqrt{3} / A$$

$$A \sin \alpha = 1 / A$$



$$A = 2 \quad \leftarrow (1)$$

$$\alpha = \pi/6 \quad \leftarrow (1)$$

$$\therefore = 2 \cos(x + \frac{\pi}{6})$$

$$(ii) 3 - 3 \cos x + \sqrt{3} \sin x$$

$$= 3 - \sqrt{3} (\sqrt{3} \cos x - \sin x)$$

$$= 3 - \sqrt{3} \left( 2 \cos(x + \frac{\pi}{6}) \right) \quad \leftarrow (1)$$

max value 2  
min value -2

$$\therefore \text{Max. value} = 3 - \sqrt{3}(-2)$$

$$= 3 + 2\sqrt{3}$$

$\leftarrow (1)$

$$b) P(x) = -x^3 + 3x + 2$$

$$(i) P(-1) = -(-1)^3 + 3(-1) + 2$$

$$= 1 - 3 + 2$$

$$= 0$$

$\therefore x+1$  is a factor (1)

(ii)

$$\begin{array}{r} -x^2 + x + 2 \quad \leftarrow (1) \\ x+1 \overline{) -x^3 + 0x^2 + 3x + 2} \\ \underline{-x^3 - x^2} \phantom{+ 2} \\ x^2 + 3x \phantom{+ 2} \\ \underline{x^2 + x} \phantom{+ 2} \\ 2x + 2 \\ \underline{2x + 2} \\ 0 \end{array}$$

$$\therefore P(x) = (x+1) \cdot -(x^2 - x - 2)$$

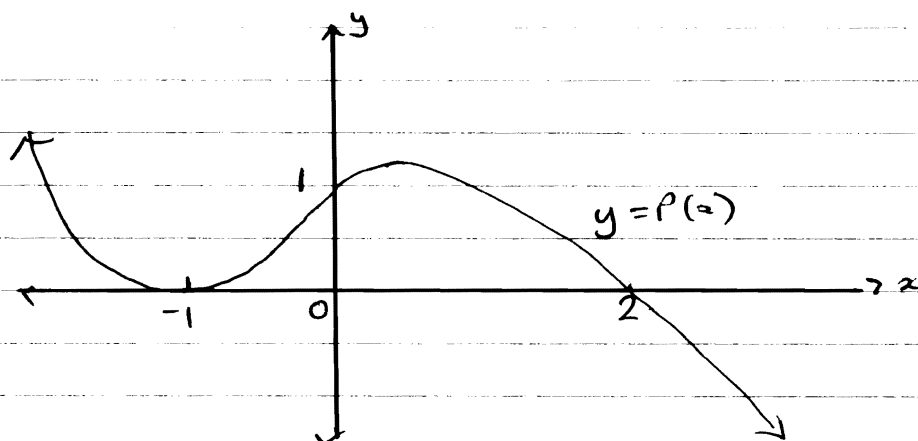
$$= -(x+1)(x+1)(x-2)$$

$$= -(x+1)^2(x-2)$$

$$= (2-x)(x+1)^2$$

$\leftarrow (1)$

(iii)



(1) Single root & decr. to right

(1) Double root at  $x = -1$

- c) Without restriction :  ${}^8C_4 = 70$  — (1)  
All men : 5  
All women ~~60~~ : not possible  
 $\therefore$  No. of ways =  $70 - 5 = 65$  — (1)

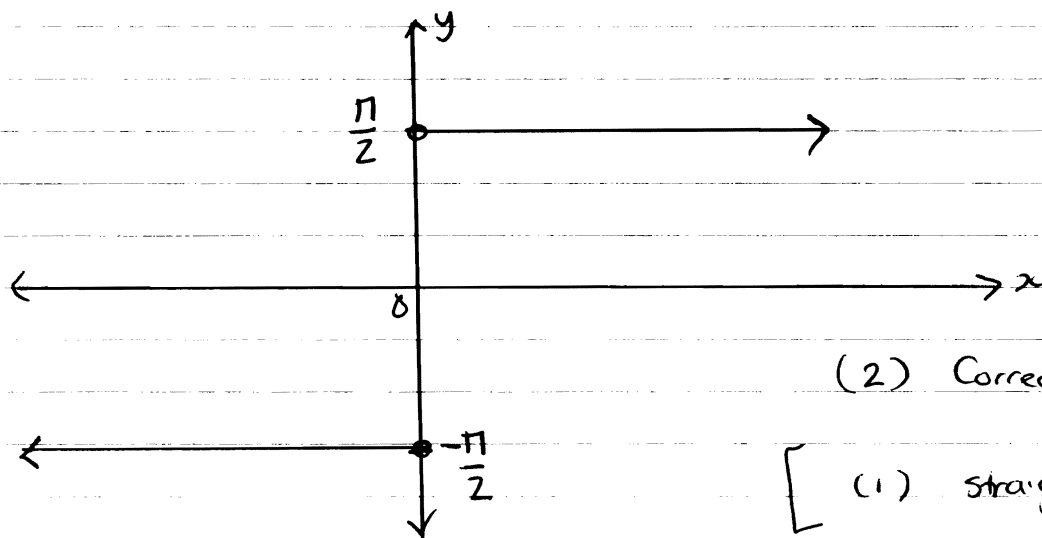
d)  $f(x) = \tan^{-1} \frac{4}{x} + \tan^{-1} \frac{x}{4}$

$$\begin{aligned} \text{(i)} \quad f'(x) &= \frac{1}{1 + \left(\frac{4}{x}\right)^2} \cdot \frac{-4}{x^2} + \frac{4}{16 + x^2} \\ &= \frac{-4}{x^2 + 16} + \frac{4}{16 + x^2} \\ &= 0 \end{aligned}$$

← (1)  
One or both correct

← (1)

(ii)



(2) Correct 2 rays

[ (1) straight line ]

Question 12.

a)  $\sin (2 \tan^{-1} \sqrt{2})$

$$= \sin ( \quad )$$

$$= 2 \sin A \cos A$$

← (1)

$$= 2 \cdot \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}$$

$$= \frac{2\sqrt{2}}{3}$$

← (1)

b) (i)  $\tan \phi = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$\left. \begin{array}{l} m_1 = 2 \\ m_2 = \frac{1}{4} \end{array} \right\} \leftarrow (1)$$

$$= \left| \frac{2 - \frac{1}{4}}{1 + 2(\frac{1}{4})} \right|$$

← (1)

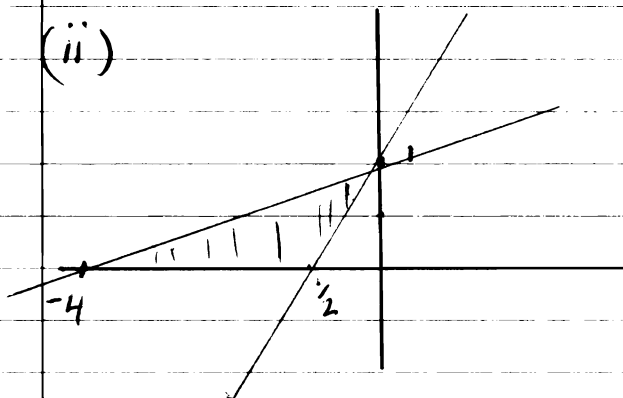
$$= \frac{1\frac{3}{4}}{1\frac{1}{2}}$$

$$= \frac{7}{6}$$

$$= \frac{7}{6}$$

$$\phi = \tan^{-1} \left( \frac{7}{6} \right) \doteq 49^\circ \quad \leftarrow (1)$$

(ii)



Intersect at (0, 1) ← (1)

$$A = \frac{1}{2} \times 3.5 \times 1$$

$$= 1.75 \text{ units}^2 \quad \leftarrow (1)$$

$$\begin{aligned}
 \text{c)} \quad & 2 \sin x \cos x - \cos x = 0 \\
 & \cos x (2 \sin x - 1) = 0 \\
 & \cos x = 0 \quad \text{or} \quad \sin x = \frac{1}{2} \\
 & \underline{x = \frac{\pi}{2}, \frac{3\pi}{2}} \quad \leftarrow (1) \qquad \underline{x = \frac{\pi}{6}, \frac{5\pi}{6}} \quad \leftarrow (1)
 \end{aligned}$$

$$\text{d)} \quad \ln(2x-3)(x+3) - \ln(2x-3)(x-2) = 2$$

$$\ln \left( \frac{(2x-3)(x+3)}{(2x-3)(x-2)} \right) = 2 \quad \leftarrow (1)$$

$$\frac{x+3}{x-2} = e^2 \quad \leftarrow (1)$$

$$x+3 = e^2 x - 2e^2$$

$$x(1-e^2) = -2e^2 - 3$$

$$\underline{x = \frac{-2e^2 - 3}{1-e^2} \quad \text{or} \quad \frac{2e^2 + 3}{e^2 - 1}} \quad (1)$$

$$\text{e)} \quad 4 \sin^3 \theta = 3 \sin \theta - \sin 3\theta$$

$$\therefore \int 2 \sin^3 \theta \cdot d\theta = \frac{1}{2} \int 3 \sin \theta - \sin 3\theta \cdot d\theta \quad \leftarrow (1)$$

$$= \frac{1}{2} \left( -3 \cos \theta + \frac{1}{3} \cos 3\theta \right) + c$$

$$= \underline{\underline{-\frac{3}{2} \cos \theta + \frac{1}{6} \cos 3\theta + c}} \quad \leftarrow (1)$$

$$\begin{aligned}
 f) & \sin(a+b) \cos\left(\frac{\pi}{2} - (a+b)\right) - \cos(a+b) \cos(\pi - (a+b)) \\
 &= \sin(a+b) \sin(a+b) - \cos(a+b) \cdot -\cos(a+b) \\
 &= \sin^2(a+b) + \cos^2(a+b) \quad \leftarrow (1) \\
 &= 1 \quad \text{--- (1)}
 \end{aligned}$$

Question 13.

a) If  $n=1$

$$\left. \begin{aligned}
 \text{LHS} &= 1^3 = 1 \\
 \text{RHS} &= \frac{1}{4} \cdot 1^2 \cdot 2^2 = \frac{1}{4}(4) = 1
 \end{aligned} \right\} \text{LHS} = \text{RHS}$$

$\therefore$  True for  $n=1$

Assume true for  $n=k$

ie Assume

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4} k^2 (k+1)^2$$

Now prove true for  $n=k+1$

ie

$$\text{Prove } \underbrace{1^3 + 2^3 + 3^3 + \dots + k^3}_{\text{LHS}} + (k+1)^3 = \frac{1}{4} (k+1)^2 (k+2)^2$$

LHS =

$$\frac{1}{4} k^2 (k+1)^2 + (k+1)^3$$

(3)

$$= \frac{1}{4} (k+1)^2 (k^2 + 4(k+1))$$

$$= \frac{1}{4} (k+1)^2 (k^2 + 4k + 4)$$

$$= \frac{1}{4} (k+1)^2 (k+2)^2$$

= RHS

$\therefore$  If true for  $n=k$ , then also true for  $n=k+1$

Now statement is true for  $n=1$

$\therefore$  Also true for  $n=2, 3, 4, \dots$  and by induction, true for all positive integers  $n$ .

$$(ii) \frac{1}{4} \cdot 50^2 \cdot 51^2 - \frac{1}{4} \cdot 10^2 \cdot 11^2$$

$$= 1622600 \quad \leftarrow (1)$$

b) (i) # Sub.  $S(0, a)$  into eqn:

$$a = \frac{p+q}{2}, 0 - apq \quad \leftarrow (1)$$

$$a = -apq \quad \therefore pq = -1$$

$$(ii) X = \left( \frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right) = \left( a(p+q), \frac{a(p^2+q^2)}{2} \right)$$

$$T = (a(p+q), -a) \quad (1) \text{ for } T$$

$$W = \left( a(p+q), \frac{\frac{a}{2}(p^2+q^2) - a}{2} \right) \quad (1) \text{ for } W$$

$$= \left( a(p+q), \frac{a(p^2+q^2-2a)}{4} \right)$$

$$\text{At } W: x = a(p+q) \quad p+q = \frac{x}{a}$$

$$y = \frac{a(p^2+q^2-2a)}{4} \quad p^2+q^2 = \frac{4y}{a} + 2$$

$$(p+q)^2 - 2pq = \frac{4y}{a} + 2$$

$$\left( \frac{x}{a} \right)^2 - 2(-1) = \frac{4y}{a} + 2$$

$$\frac{x^2}{a^2} + 2 = \frac{4y}{a} + 2$$

$$\text{Locus: } \underline{x^2 = 4ay} \quad (1) \text{ for locus}$$

c)  $T = P + Ae^{kt}$

(i)  $\frac{dT}{dt} = \frac{d}{dt} (P + Ae^{kt})$

$= 0 + Ae^{kt} \cdot k$

$= k(T - P)$  but  $Ae^{kt} = T - P$  (1)  
as required.

(ii)  $P = 23$  :  $T = 23 + Ae^{kt}$

$t = 0, T = 100$  :  $100 = 23 + Ae^{k(0)}$   $\therefore A = 77$  — (1)

$t = 2, T = 93$   $93 = 23 + 77e^{2k}$

$70 = 77e^{2k}$

$\frac{70}{77} = e^{2k}$

$k = \frac{1}{2} \ln \frac{70}{77} \doteq -0.0477$  — (1)

(iii)  $T = 23 + 77e^{-0.0477t}$

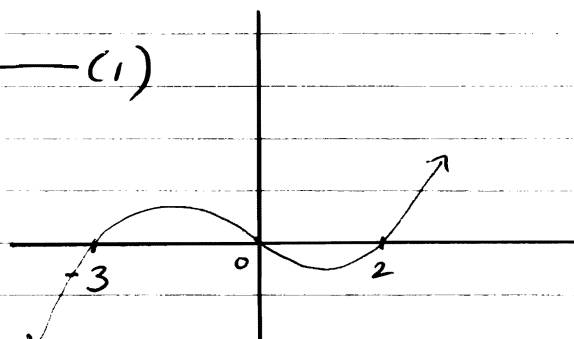
$80 = 23 + 77e^{-0.0477t}$

$\frac{57}{77} = e^{-0.0477t}$

$t = \frac{-1}{0.0477} \ln \frac{57}{77} \doteq 6.3 \text{ min}$  — (1)

d)  $\left(\frac{x^2+x-6}{x}\right)^{xx^2} \geq 0$  (note:  $x \neq 0$ )

$x(x+3)(x-2) \geq 0$  — (1)



Soln :  $\underline{-3 \leq x < 0}$  ,  $\underline{x \geq 2}$   
 $\swarrow$  (1)  $\nwarrow$  (1)

Q14

$$a) \int \tan^2 2x \cdot dx = \int \sec^2 2x - 1 \cdot dx \quad (1)$$

$$= \frac{1}{2} \tan 2x - x + \underline{\underline{c}} \quad (1) \text{ must have } +c.$$

$$b) \frac{2^{4n} \times 3^{2n}}{8^n \times 6^n} + 3^n$$

$$= \frac{2^{4n} \times 3^{2n}}{2^{3n} \times 2^n \times 3^n} + 3^n \quad (1)$$

$$= \frac{2^{4n} \times 3^{2n}}{2^{4n} \times 3^n} + 3^n$$

$$= 3^n + 3^n$$

$$= 2 \times 3^n \quad (1)$$

$$c) i) f: y = x - \frac{1}{x} \quad (x > 0)$$

$$f^{-1}: x = y - \frac{1}{y} \quad (y > 0)$$

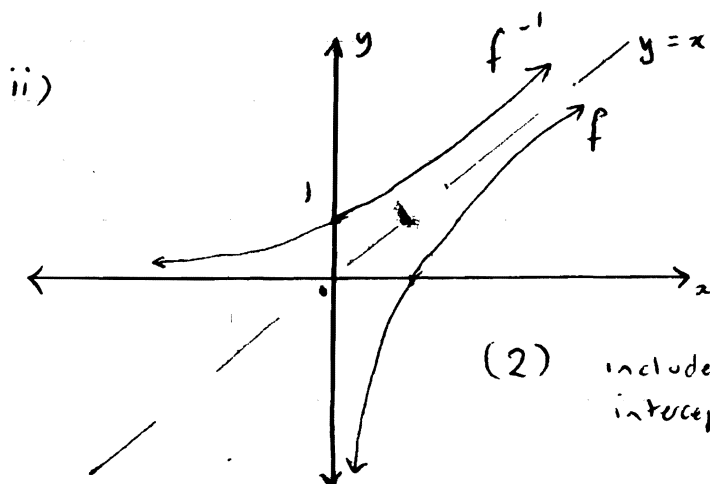
$$\begin{aligned} xy &= y^2 - 1 \\ y^2 - xy - 1 &= 0 \\ y &= \frac{x \pm \sqrt{(-x)^2 - 4 \cdot 1 \cdot -1}}{2} = \frac{x \pm \sqrt{x^2 + 4}}{2} \end{aligned} \quad (1)$$

$$= \frac{x}{2} \pm \sqrt{\frac{x^2 + 4}{4}}$$

Some explanation  
↓ needed

$$\therefore y = \frac{x}{2} + \sqrt{\frac{x^2}{4} + 1} \quad (1)$$

because since  $x > 0$  in  $f(x)$ , then  $y > 0$  in  $f^{-1}(x)$   
+ (not  $\pm$ ).

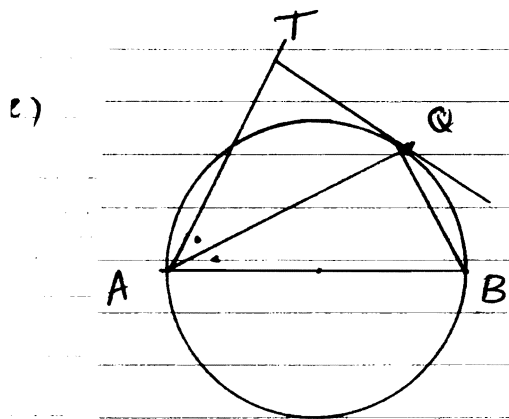


(2) include intercepts.

(iii) D: all real  $x$  (1)

R:  $y > 0$  (1)

$$\begin{aligned}
 1) \quad LHS &= \frac{\sin 8x}{1 + \cos 8x} = \frac{2 \sin 4x \cos 4x}{1 + 2\cos^2 4x - 1} & (1) \\
 &= \frac{\cancel{2} \sin 4x \cos \cancel{4x}}{\cancel{2} \cos^2 4x} \\
 &= \frac{\sin 4x}{\cos 4x} & (1) \\
 &= \tan 4x \\
 &= RHS
 \end{aligned}$$



Join BQ.

In  $\triangle TQA$ ,  $\triangle QBA$

$$\angle TQA = \angle QBA \quad (\text{alternate segment theorem}) \quad (1)$$

$$\angle TAQ = \angle QAB \quad (QA \text{ bisects } \angle TAB)$$

$$\therefore \triangle TQA \parallel \triangle QBA \quad (\text{matching } \angle\text{s equal}) \text{ or (AA)}$$

$$\therefore \angle QTA = \angle BQA \quad (\text{matching } \angle\text{s of similar } \Delta\text{s are equal}) \quad (1)$$

$$\text{but } \angle BQA = 90^\circ \quad (\angle \text{ in semicircle}) \quad (1)$$

$$\therefore \angle QTA = 90^\circ$$